EXPERIMENT NO: 01

Implementation of Java Programming Fundamentals Using a Student Utility Program

1. Title of the Experiment

Implementation of Java Programming Fundamentals Using a Student Utility Program

2. Mapped Outcomes

Course Outcome (CO): CO2
Program Outcomes (POs): PO1, PO5, PO9, PO10

3. Aim

To design and implement a menu-driven Student Utility Program in Java demonstrating variables, data types, operators, input/output, conditional statements, looping statements, and parameterized user-defined methods.

4. Algorithm / Design

1. Start the program.
2. Import the Scanner class for user input.
3. Create the StudentUtilityProgram class.
4. Define methods for: Factorial, Prime number checking, Maximum of two numbers, Area of a circle.
5. Display the Student Utility Program menu.
6. Read the user's choice.
7. Use a switch statement to execute the selected operation.
8. Perform student information, conditional, looping, and method operations.
9. Repeat the menu until the user selects 0.
10. Close the Scanner and terminate the program.

5. Java Program

Package: task1   Class: StudentUtilityProgram

```java
package task1;

import java.util.Scanner;

public class StudentUtilityProgram {

    // ---------- Part D : Methods ----------

    // Factorial Method
    public static long factorial(int n) {
        long fact = 1;

        for (int i = 1; i <= n; i++) {
            fact *= i;
        }

        return fact;
    }

    // Prime Number Method
    public static boolean isPrime(int n) {

        if (n <= 1)
```

```java
        return false;

    for (int i = 2; i <= Math.sqrt(n); i++) {

        if (n % i == 0)
            return false;
    }

    return true;
}

// Maximum of Two Numbers
public static int max(int a, int b) {
    return (a > b) ? a : b;
}

// Area of Circle
public static double areaOfCircle(double radius) {
    return Math.PI * radius * radius;
}

// ---------- Main Method ----------
public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    int choice;

    do {

        System.out.println("\n===== STUDENT UTILITY PROGRAM =====");
        System.out.println("1. Student Information");
        System.out.println("2. Even or Odd");
        System.out.println("3. Largest of Three Numbers");
        System.out.println("4. Display Grade");
        System.out.println("5. Day of the Week");
        System.out.println("6. Multiplication Table");
        System.out.println("7. Numbers from 1 to N");
        System.out.println("8. Sum of First N Natural Numbers");
        System.out.println("9. Fibonacci Series");
        System.out.println("10. Factorial");
        System.out.println("11. Prime Number Check");
        System.out.println("12. Maximum of Two Numbers");
        System.out.println("13. Area of Circle");
        System.out.println("0. Exit");

        System.out.print("Enter your choice: ");

        choice = sc.nextInt();

        switch (choice) {

            // ---------- Part A ----------
            case 1:

                sc.nextLine();

                System.out.print("Enter Student Name: ");
                String name = sc.nextLine();

                System.out.print("Enter Roll Number: ");
                String roll = sc.nextLine();

                System.out.print("Enter Marks in Subject 1: ");
                int m1 = sc.nextInt();

                System.out.print("Enter Marks in Subject 2: ");
                int m2 = sc.nextInt();

                System.out.print("Enter Marks in Subject 3: ");
                int m3 = sc.nextInt();

                int total = m1 + m2 + m3;

                double percentage = total / 3.0;

                System.out.println("\n----- Student Details -----");
                System.out.println("Name : " + name);
                System.out.println("Roll Number : " + roll);
                System.out.println("Total Marks : " + total);

                System.out.printf(
                    "Percentage : %.2f%%\n",
                    percentage
                );
                break;
```

```java
// ---------- Part B ----------
case 2:

    System.out.print("Enter a number: ");

    int num = sc.nextInt();

    if (num % 2 == 0)
        System.out.println("Even Number");
    else
        System.out.println("Odd Number");

    break;


case 3:

    System.out.print("Enter three numbers: ");

    int a = sc.nextInt();
    int b = sc.nextInt();
    int c = sc.nextInt();

    int largest = Math.max(a, Math.max(b, c));

    System.out.println(
        "Largest Number = " + largest
    );

    break;


case 4:

    System.out.print("Enter Percentage: ");

    double per = sc.nextDouble();

    if (per >= 90)
        System.out.println("Grade A");

    else if (per >= 75)
        System.out.println("Grade B");

    else if (per >= 60)
        System.out.println("Grade C");

    else if (per >= 40)
        System.out.println("Grade D");

    else
        System.out.println("Fail");

    break;


case 5:

    System.out.print(
        "Enter Day Number (1-7): "
    );

    int day = sc.nextInt();

    switch (day) {

        case 1:
            System.out.println("Monday");
            break;

        case 2:
            System.out.println("Tuesday");
            break;

        case 3:
            System.out.println("Wednesday");
            break;

        case 4:
            System.out.println("Thursday");
            break;

        case 5:
            System.out.println("Friday");
            break;
        case 6:
```

```
            System.out.println("Saturday");
            break;

        case 7:
            System.out.println("Sunday");
            break;

        default:
            System.out.println("Invalid Day");
    }

    break;


// ---------- Part C ----------
case 6:

    System.out.print("Enter Number: ");

    int table = sc.nextInt();

    for (int i = 1; i <= 10; i++) {

        System.out.println(
            table + " x " + i +
            " = " + (table * i)
        );
    }

    break;


case 7:

    System.out.print("Enter N: ");

    int n = sc.nextInt();

    for (int i = 1; i <= n; i++) {
        System.out.print(i + " ");
    }

    System.out.println();

    break;


case 8:

    System.out.print("Enter N: ");

    int num1 = sc.nextInt();

    int sum = 0;

    for (int i = 1; i <= num1; i++) {
        sum += i;
    }

    System.out.println("Sum = " + sum);

    break;


case 9:

    System.out.print(
        "Enter Number of Terms: "
    );

    int terms = sc.nextInt();

    int first = 0;
    int second = 1;

    System.out.print(
        "Fibonacci Series: "
    );

    for (int i = 1; i <= terms; i++) {

        System.out.print(first + " ");

        int next = first + second;

        first = second;
```

```
                second = next;
            }

            System.out.println();

            break;


        // ---------- Part D ----------
        case 10:

            System.out.print("Enter Number: ");

            int factNum = sc.nextInt();

            System.out.println(
                "Factorial = " +
                factorial(factNum)
            );

            break;


        case 11:

            System.out.print("Enter Number: ");

            int primeNum = sc.nextInt();

            if (isPrime(primeNum))
                System.out.println("Prime Number");
            else
                System.out.println(
                    "Not a Prime Number"
                );

            break;


        case 12:

            System.out.print(
                "Enter Two Numbers: "
            );

            int x = sc.nextInt();
            int y = sc.nextInt();

            System.out.println(
                "Maximum = " + max(x, y)
            );

            break;


        case 13:

            System.out.print("Enter Radius: ");

            double radius = sc.nextDouble();

            System.out.printf(
                "Area = %.2f\n",
                areaOfCircle(radius)
            );

            break;


        // ---------- Exit ----------
        case 0:

            System.out.println(
                "Program Terminated."
            );

            break;


        default:

            System.out.println(
                "Invalid Choice!"
            );
    }
} while (choice != 0);
```

```
            sc.close();
        }
    }
```

6. Test Cases & Execution Data

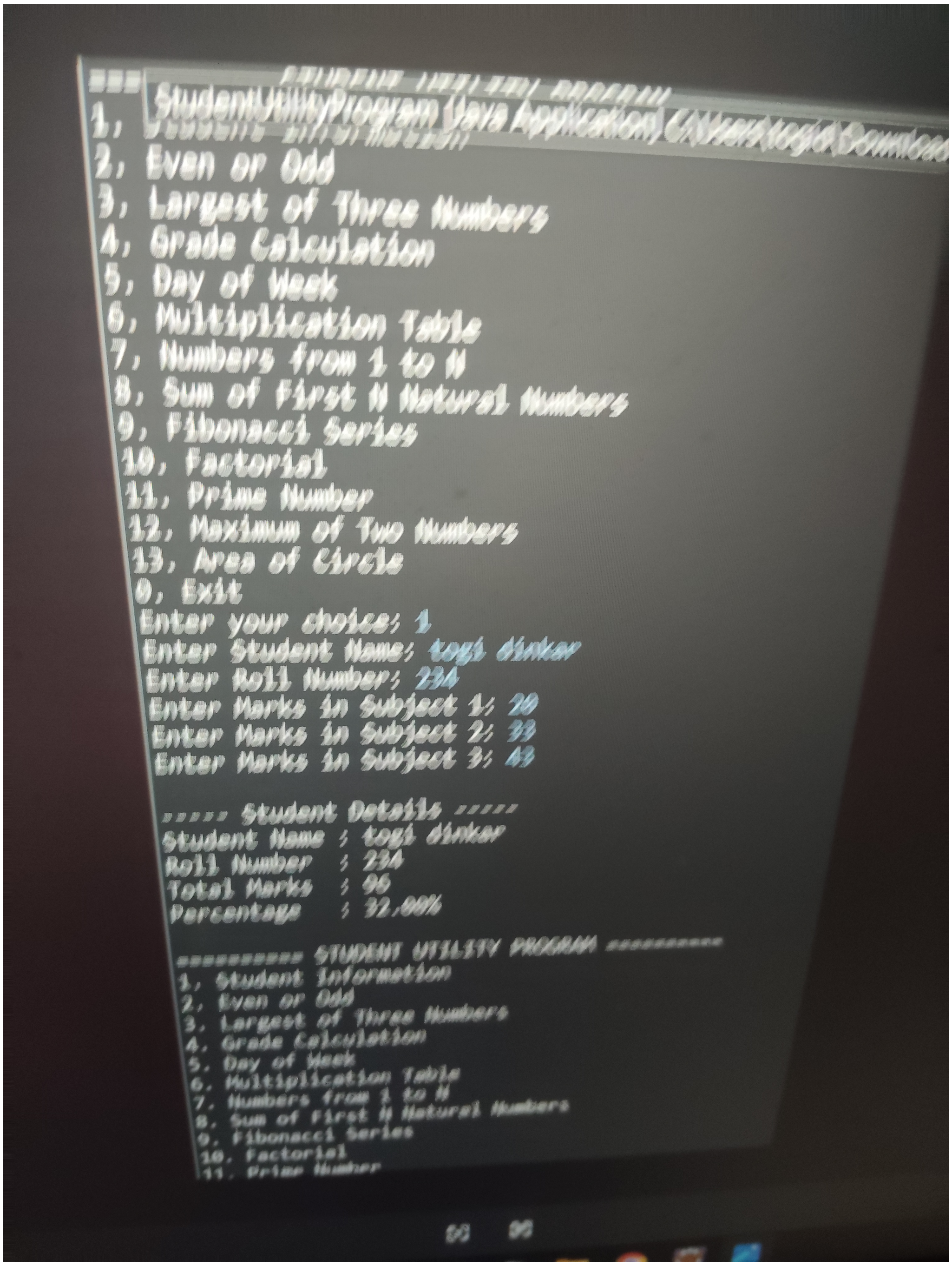
Test Case	Input	Expected Output	Status
TC-01: Student Information	Name = Dinkar Roll Number = 101 Marks = 80 85 90	Total Marks : 255 Percentage : 85.00%	Passed
TC-02: Even or Odd	25	Odd Number	Passed
TC-03: Largest of Three Numbers	25 78 45	Largest Number = 78	Passed
TC-04: Grade	85	Grade B	Passed
TC-05: Fibonacci Series	7	Fibonacci Series: 0 1 1 2 3 5 8	Passed
TC-06: Factorial	5	Factorial = 120	Passed
TC-07: Prime Number	17	Prime Number	Passed
TC-08: Area of Circle	7	Area = 153.94	Passed

7. Output Proof

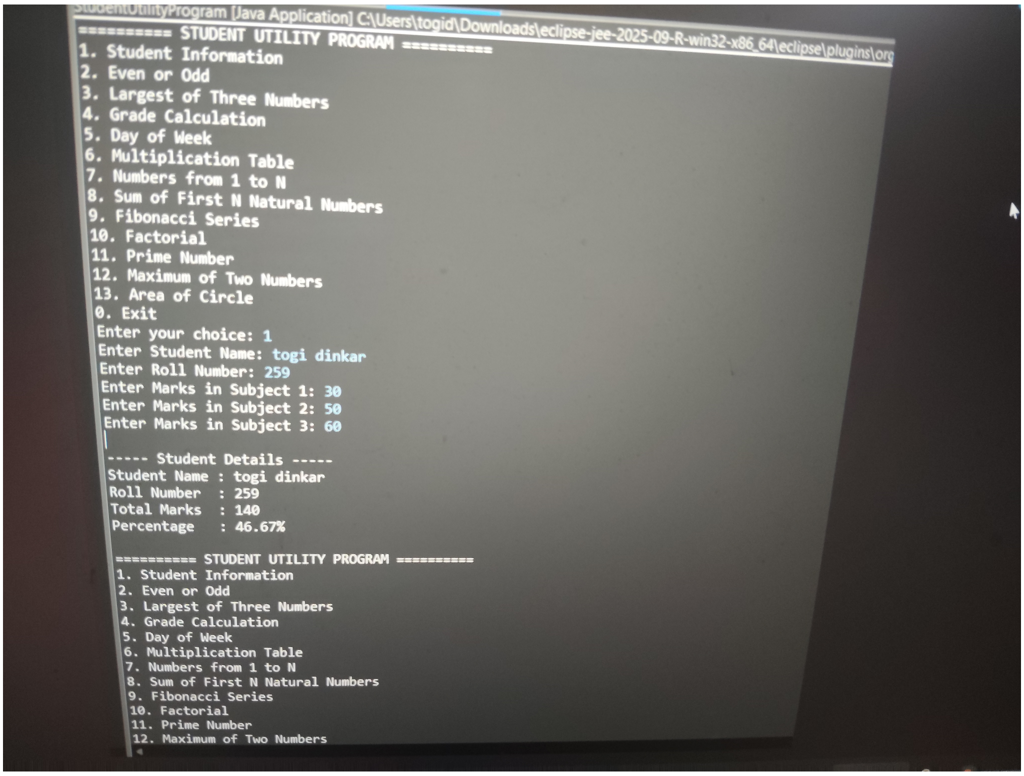
The program was successfully executed using Eclipse IDE. Console screenshots showing the program menu and execution results are attached below.

Example:

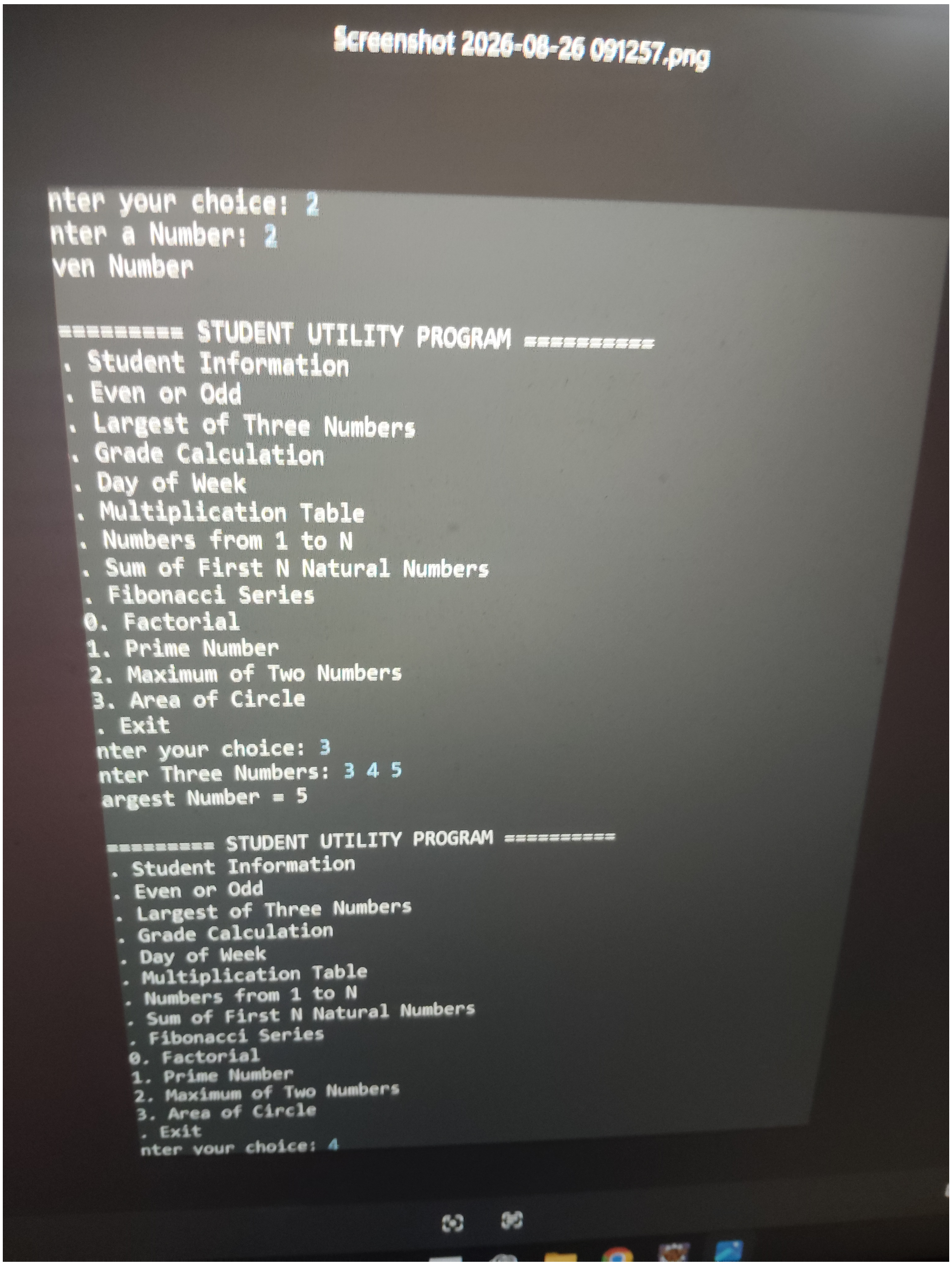
```
===== STUDENT UTILITY PROGRAM =====
1. Student Information
2. Even or Odd
3. Largest of Three Numbers
4. Display Grade
5. Day of the Week
6. Multiplication Table
7. Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Prime Number Check
12. Maximum of Two Numbers
13. Area of Circle
0. Exit

Enter your choice: 10
Enter Number: 5
Factorial = 120
```



Screenshot 1: Eclipse Console Output



Screenshot 2: Eclipse Console Output



Screenshot 3: Eclipse Console Output

8. Learning Outcome / Result

Successfully implemented and verified the Student Utility Program using Java Programming Fundamentals. The experiment demonstrated variables, data types, operators, input/output, conditional statements, switch statements, loops, and parameterized user-defined methods. The menu-driven program provided practical understanding of Java programming and control flow.